



**BALL STATE**  
UNIVERSITY

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# **Module 13 Lecture - F Distribution and One-Way ANOVA**

Introduction to Statistical Methods

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# 1 Overview and Introduction

## 1.1 Textbook Learning Objectives

- Interpret the F probability distribution as the number of groups and the sample size change.
- Discuss two uses for the F distribution: one-way ANOVA and the test of two variances.
- Conduct and interpret one-way ANOVA.
- Conduct and interpret hypothesis tests of two variances.

## 1.2 Instructor Learning Objectives

- Appreciate the ANOVA as another test in your toolbox alongside the other tests for dealing with categorical and continuous outcome variables
- Understand the F-distribution as another practical distribution for deriving inferences and conclusions

### 1.3 Introduction

## 2 One-way ANOVA

### 2.1 Introduction

## 3 The F Distribution and the F-Ratio

### 3.1 Introduction

## 4 Facts About the F Distribution

### 4.1 Introduction

## 5 Test of Two Variances

### 5.1 Introduction

## 6 Conclusion

### 6.1 Recap

*The instructor-provided glossary may not include all terms worth memorizing, make sure you consider using the vocabulary list in your book and your own judgment to make sure you have all relevant terms*